WORKED SOLUTION

STEP II 1999, Question 14

Probability and Statistics • Optimal Stopping

STEP II 1999, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s2-q14

The key idea

The die has no memory. Whenever you decide to throw again, you are back at the very start of the same game, so the best you can hope for from that point on is the same number V, the best expected score of the whole game. That turns the decision after each throw into a simple comparison: keep the number you have if it beats V, otherwise throw again. Writing this down gives one equation for V, and solving it tells you both the value and the strategy.

Solution

Setting up

Call a throw that shows 1 *forced*: the game ends there with score 1. After any other throw $x \in \{2,3,4,5,6\}$ we choose to stop (score x) or continue. Every score lies between 1 and 6, so every strategy has an expected score in $[1,6]$. Let

$$V = \text{the largest expected score that any strategy can achieve}$$

(strictly, the least upper bound over all strategies; we'll see that it is attained).

A strategy that scores 4

Consider the strategy: *stop as soon as you throw* 4, 5 *or* 6 (and, of course, when you throw 1). Let E be its expected score. Condition on the first throw. With probability $\frac{1}{6}$ each, it shows $1,4,5,6$ and the game ends with that score. With probability $\frac{2}{6}$ it shows 2 or 3; we throw again and, because the die has no memory, the expected score from there is E again. So

$$E = \tfrac{1}{6}(1+4+5+6) + \tfrac{2}{6}E \quad\Longrightarrow\quad \tfrac{4}{6}E = \tfrac{16}{6} \quad\Longrightarrow\quad E = 4.$$

(The game ends with probability 1, since the chance of n throws without stopping is $(\frac{1}{3})^n \to 0$.) Hence $V \geq 4$.

No strategy can beat 4

Take any strategy at all; it may even depend on the whole history of throws. Look at its first throw x.

- If $x = 1$ the score is 1.
- If $x \geq 2$ and the strategy stops, the score is x.
- If $x \geq 2$ and the strategy continues, then what it does from now on is just some strategy for a fresh copy of the game (the earlier throws can't affect later ones). So its expected score from here is at most V.

In every case the expected score given $x \geq 2$ is at most $\max(x, V)$. Hence every strategy's expected score is at most

$$g(V) = \tfrac{1}{6}\Big(1 + \sum_{x=2}^{6} \max(x, V)\Big),$$

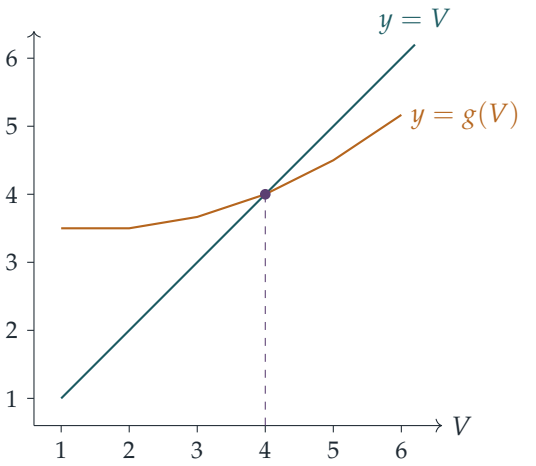
and taking the least upper bound over strategies gives $V \leq g(V)$.

? **Ask yourself** How does the inequality $V \leq g(V)$ pin V down?

Look at $h(V) = g(V) - V$. Each term $\max(x, V)$ rises at rate 0 or 1 as V increases, and there are only five of them, so g rises at rate at most $\frac{5}{6}$. Hence h is strictly decreasing. At $V = 4$,

$$g(4) = \tfrac{1}{6}(1 + 4 + 4 + 4 + 4 + 5 + 6) = \tfrac{24}{6} = 4,$$

so $h(4) = 0$, and $h(V) < 0$ for every $V > 4$. Since $h(V) \geq 0$, we must have $V \leq 4$.



Conclusion

Combining the two parts, $V = 4$: the best possible expected score is 4, and it is achieved by stopping on 4, 5 or 6 and throwing again on 2 or 3. This is exactly the rule "stop if

your throw is at least V": a 2 or 3 is worse than the 4 you expect by carrying on, while a 5 or 6 is better.

A 4 is a tie. Stopping on 4 gives 4, and continuing also gives expected score 4. So "stop only on 5 or 6" is equally good: by the same conditioning,

$$E' = \tfrac{1}{6}(1+5+6) + \tfrac{3}{6}E' \quad \Longrightarrow \quad E' = 4.$$

Any other rule does strictly worse; for instance stopping on 3 or more gives $\frac{1+3+4+5+6}{5} = \frac{19}{5}$, and stopping only on 6 gives $\frac{1+6}{2} = \frac{7}{2}$.

Checks

A neat shortcut for any rule "stop on 1 and on the faces in a set S": each throw ends the game with the same probability, and the face it ends on is equally likely to be any of $\{1\} \cup S$, so the expected score is the mean of those faces. For $S = \{4,5,6\}$ that is $\frac{16}{4} = 4$; for $S = \{5,6\}$ it is $\frac{12}{3} = 4$. We checked all 32 such rules exactly: only these two give 4, and the next best is $\frac{19}{5}$. A simulation of 400 000 games stopping on 4 or more gave an average of 3.998.

Common slips

- Using 3.5, the mean of one throw, as the value of continuing. Continuing is worth more than 3.5 because you keep the option to throw again, so "stop on 4 or more" isn't justified by comparing with 3.5.
- Forgetting that a 1 is forced, and so writing the continuation value as if you could throw forever without risk.
- Checking only threshold strategies without explaining why a cleverer, history-dependent rule can't do better. The memorylessness argument above handles every strategy.

Beyond STEP

This is a small example of *optimal stopping*, and the equation $V = g(V)$ is a *Bellman equation*, the basic tool of dynamic programming and Markov decision processes. The same reasoning prices American options in finance and solves the famous secretary problem. Grimmett and Welsh, *Probability: An Introduction*, covers the conditioning arguments used here, and Blitzstein and Hwang, *Introduction to Probability*, has many problems in the same spirit.

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